Gabriel Kassie

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SUMMARY

Computer engineering student with experience in technical projects for Solar Devils and Destination Express LLC. Seeking internship during the summer of 2026 in embedded systems, hardware engineering, software development, and related areas.

EDUCATION

B.S.E., Computer Engineering May 2028
Arizona State University, Tempe, AZ 3.61 GPA
Barrett, the Honors College
Relevant Coursework: Circuits I, Object-Oriented Programming and Data Structures, Computer Organization and Assembly, Embedded Microprocessor Systems

SKILLS

Hardware and Embedded: KiCAD, LTspice, ARM Cortex-M0+ (NXP FRDM-KL46Z), RP2350-CAN
Programming Languages: C, C++, Assembly (MIPS), Python, MATLAB, Java
Web Development and Tools: HTML, CSS, JavaScript, Django, MySQL, Unix/Linux, Git/GitHub, VS Code

RESEARCH AND ACADEMIC PROJECTS

Embedded Microprocessor Systems: Autonomous Robot Lab Project January 2026 – May 2026

- Accomplished 100% mission success and environmental adaptation across multiple surfaces by constructing a bare-metal vehicle platform on the ARM Cortex-M0+ (FRDM-KL46Z) utilizing a dual-motor chassis and a servo
- Navigated real-time trajectories and obstacle avoidance within 2cm path accuracy by programming a PID control system and logic to interface with RGB color, line, and ultrasonic sensors via I2C, ADC, and PWM
- Verified hardware-software reliability and signal integrity by implementing resistor-based voltage scaling for processor protection and utilizing oscilloscopes to troubleshoot sub-millisecond pulse timing

Solar Devils, ASU Solar Car Team: Electrical Integrations Engineer August 2024 – Present

- Architecting C-based firmware for the RP2350-CAN board to implement safety-critical throttle control logic
- Optimized vehicle-wide electrical systems by designing comprehensive electrical schematics in KiCAD to integrate power distribution and communication protocols
- Enhanced real-time vehicle diagnostics by decoding 700+ CAN Bus data points utilizing Python and Kivy to develop a driver-facing dashboard for critical system diagnostics

Engineering Projects in Community Service, ASU: Prototyping Lead January 2025 – May 2025

- Developed an Arduino-based system to monitor pH, Dissolved Oxygen, and ORP levels
- Applied manual calibration offsets and formulas to convert raw sensor voltages into units for monitoring, improving sensor accuracy by 20% compared to baseline readings
- Collaborated with four engineers to refine system specifications and implement real-time data transmission from Arduino to a mobile application

WORK EXPERIENCE

Destination Express LLC, **Phoenix, AZ: Software Engineer** December 2024 – January 2025

- Streamlined auto parts delivery logistics for 60+ users by managing 1000+ orders through the deployment of a full-stack web application with Python, Django, and MySQL
- Developed and integrated multi-level dashboards for shippers, drivers, and dispatchers using HTML, CSS, and JavaScript, incorporating secure authentication, real-time order tracking, and management
- Built comprehensive customer and courier features, including new order forms, manifest generation with print functionality, and request handling for quotes, cancellations, and courier applications